CORROSION OF IRON

When a metal is attacked by air, water or other substances, the metal is said to **corrode**. Usually the more reactive a metal is, the more readily it corrodes. The most common example of corrosion is the rusting of iron. Corrosion is an **oxidation** process. Remember, oxidation involves any (or all) of the following:

- addition of oxygen
- loss of electrons
- loss of hydrogen.

1 *Reduction* is the opposite of oxidation. Outline what reduction involves.

Corrosion is a process that costs Australia billions of dollars each year. Steel, an alloy of iron, is a commonly used structural material; it is strong, easily shaped and reasonably cheap. However, it rusts in damp air.

2 Define *alloy*.

3 Give two structural uses of steel.

When iron corrodes, rust forms in flakes. Damp air can get past the flakes and attack the metal below. Given enough time, a piece of iron can rust all the way through.

There are many ways to protect iron from rusting. They often involve coating the iron with paint or grease to keep out air and water. **Galvanising** is a process that covers iron (or steel) with a coating of zinc. Zinc is more reactive than iron, so will corrode in preference to the iron even if the zinc coating is scratched. Rust is hydrated iron oxide and has the formula $Fe_2O_3.xH_2O$, where x can be a number between one and three.

4 What is the oxidation state of iron in hydrated iron oxide?

Rust forms in a series of reactions. The first step is the oxidation of iron:

$$Fe\ (s) \rightarrow Fe^{2+}\ (aq) + 2e^-$$

The electrons lost in this reaction are used by oxygen (which is reduced):

$$\tfrac{1}{2}O_2\ (aq) + H_2O\ (l) + 2e^- \rightarrow 2OH^-\ (aq)$$

5 Add the two previous half-reactions to show the formation of iron (II) hydroxide, $Fe(OH)_2$.

The iron (II) hydroxide formed is insoluble. This in turn reacts with further oxygen and moisture to form iron (III) hydroxide, a reddish-brown solid.

6 Copy and balance the equation below which shows the process described above.

$$Fe(OH)_2\ (s) + O_2\ (aq) + H_2O\ (l) \rightarrow Fe(OH)_3\ (s)$$

In air, the iron (III) hydroxide dehydrates to form rust: $Fe_2O_3.xH_2O$.

7 What is meant by *dehydrates*?

The wet corrosion of iron involves the setting up of electrochemical cells on the surface of the iron, where **anodes** and **cathodes** are formed.

An anode:

- is where iron loses electrons and is oxidised
- has lower oxygen concentration than the cathode.

8 A *cathode* can be regarded as the opposite of an anode. Give the characteristics of an anode.

9 Give three desirable properties of steel.

10 Give an undesirable property of steel.

11 Define *corrode*.

12 Why can coating iron with paint stop (or slow) corrosion?

13 Define *galvanising*.

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